

MSc Proposal Defence Presentation

ANALYSIS OF PHYSICAL NOISE EFFECTS ON CORRECTNESS AND EFFICIENCY OF QUANTUM ALGORITHMS

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Speech

Good morning chairperson, panel members, and fellow students. My name is **Elisha Kidenge**, registration number **SM24/11854/24**, a Master of Science student in Computer Science at **Egerton University**.

Today I will present my research proposal titled “**Analysis of Physical Noise Effects on Correctness and Efficiency of Quantum Algorithms.**”

The research is supervised by **Dr. Nicasio Maguu** from the Department of Electrical and Control Engineering and **Dr. Duke Oeba** from the Department of Physics.

SLIDE 2 — Background of the Study

On Slide

- Quantum computing uses **qubits**
- Enables **superposition and entanglement**
- Promises exponential speedups
- Real quantum systems suffer **physical noise**
- Noise affects reliability and performance

Speech

Quantum computing represents a new paradigm of computation that uses **quantum bits, or qubits**, instead of classical bits.

Unlike classical bits, qubits exploit quantum mechanical properties such as **superposition and entanglement**, enabling certain algorithms to achieve **significant speed advantages** over classical algorithms.

However, current quantum computers operate in **noisy environments**. Physical disturbances such as **thermal fluctuations, electromagnetic interference, and decoherence** introduce errors during quantum operations.

These noise effects threaten both the **correctness of computational results** and the **efficiency of quantum algorithms**, making it essential to study how noise impacts algorithmic performance.

SLIDE 3 — Problem Statement

On Slide

- Quantum algorithms assume **ideal qubits**
- Real hardware suffers **noise and decoherence**
- Errors accumulate during computation
- Limited understanding of noise impact on:
 - Algorithm correctness
 - Computational efficiency

Speech

Most theoretical quantum algorithms are designed assuming **perfect quantum hardware**, where qubits operate without errors.

However, real quantum devices are subject to **physical noise**, which leads to **decoherence and operational errors**.

As quantum circuits grow larger, these errors **accumulate**, reducing both the **accuracy of outputs** and the **speed advantages expected from quantum algorithms**.

Therefore, there is a need to **systematically analyze how physical noise influences both correctness and efficiency in quantum algorithms**.

SLIDE 4 — Research Objectives

On Slide

General Objective

To analyze the effects of physical noise on the correctness and efficiency of quantum algorithms.

Specific Objectives

1. Identify major **sources of physical noise** in quantum systems.
2. Evaluate how noise affects **algorithm correctness**.

3. Analyze the effect of noise on **algorithm efficiency**.
4. Simulate noisy quantum circuits to measure performance degradation.

Speech

The **general objective** of this study is to analyze how physical noise affects both the **correctness and efficiency of quantum algorithms**.

The study will specifically:

First, identify the **major sources of physical noise** in quantum computing systems.

Second, evaluate how these noise sources affect the **accuracy of algorithm outputs**.

Third, analyze how noise influences the **efficiency and computational cost** of quantum algorithms.

Finally, the research will use **quantum simulation tools** to experimentally evaluate algorithm performance under noisy conditions.

SLIDE 5 — Research Questions

On Slide

1. What are the major sources of physical noise in quantum systems?
2. How does noise affect correctness of quantum algorithms?
3. How does noise impact algorithm efficiency?
4. How can simulations help evaluate noise effects?

Speech

The study will be guided by four research questions.

First, what are the **major physical noise sources** affecting quantum systems?

Second, how do these noise sources influence the **correctness of algorithm outputs**?

Third, how does noise affect the **efficiency of quantum computations**?

Finally, how can **quantum circuit simulations** be used to analyze these effects?

SLIDE 6 — Significance of the Study

On Slide

- Improves reliability of quantum algorithms
- Supports development of **error mitigation techniques**

- Contributes to **quantum algorithm design**
- Useful for **quantum computing research community**

Speech

This study is significant because it contributes to the understanding of how **real-world physical noise impacts quantum computations**.

The findings will help researchers develop **more robust quantum algorithms** and **better error mitigation techniques**.

Additionally, the study contributes to the broader field of **quantum computing research**, which is essential for the future development of practical quantum technologies.

SLIDE 7 — Scope of the Study

On Slide

Focus on:

- Selected quantum algorithms
- Simulation-based experiments
- Noise models:
 - Depolarizing noise
 - Amplitude damping
 - Phase damping

Tools:

- Quantum simulators

Speech

The study focuses on analyzing **selected quantum algorithms** under the influence of simulated noise.

Rather than using physical quantum hardware, the research will use **quantum simulators**, which allow controlled experimentation with different noise models.

The study will primarily consider **three common noise models**: depolarizing noise, amplitude damping, and phase damping.

SLIDE 8 — Theoretical Framework

On Slide

- Quantum Information Theory
- Noise models
- Quantum error theory
- Performance metrics

Speech

The research is guided by principles from **quantum information theory**, which describes how information is stored and processed in quantum systems.

The study also relies on **quantum noise models**, which mathematically represent the interaction between qubits and their environment.

Additionally, concepts from **quantum error theory** will help analyze how noise propagates through quantum circuits.

SLIDE 9 — Methodology

On Slide

Research Design: Simulation-based experimental study

Steps:

1. Select quantum algorithms
2. Implement circuits
3. Introduce noise models
4. Run simulations
5. Evaluate performance

Speech

This research adopts a **simulation-based experimental design**.

First, selected quantum algorithms will be implemented using a quantum computing framework.

Next, different **noise models** will be introduced into the quantum circuits.

Simulations will then be run under varying noise levels.

Finally, the outputs will be analyzed to determine the impact of noise on **correctness and computational efficiency**.

SLIDE 10 — Data Analysis

On Slide

Metrics:

- Accuracy
- Fidelity
- Error rate
- Execution efficiency

Speech

The simulation results will be analyzed using several **performance metrics**.

These include **accuracy**, which measures how close the output is to the expected result.

Fidelity, which measures similarity between quantum states.

Error rates, which quantify the frequency of incorrect outcomes.

And **execution efficiency**, which evaluates computational performance under noise.

SLIDE 11 — Expected Results

On Slide

- Noise reduces algorithm accuracy
- Efficiency decreases as circuit depth increases
- Certain algorithms more resilient to noise
- Insights for error mitigation

Speech

The expected results of this study are that **noise will significantly reduce the accuracy of quantum algorithms**, particularly as circuit depth increases.

The research may also reveal that **some algorithms are more resilient to noise than others**.

These findings will provide valuable insights into **designing noise-resistant quantum algorithms**.

SLIDE 12 — Work Plan

On Slide

Table showing timeline

Literature

Methodology

Simulation

Data

Report Writing

Review

Implementation

Experiments

Analysis

Speech

The research will be conducted in several stages, beginning with an extensive **literature review**, followed by **methodology implementation**.

The next stage will involve **simulation experiments**, after which the results will be analyzed.

Finally, the study findings will be compiled into the final thesis.

SLIDE 13 — Conclusion

On Slide

- Quantum noise is a major challenge
- Understanding noise impact is essential
- Research contributes to reliable quantum computing

Speech

In conclusion, **physical noise remains one of the most significant obstacles to reliable quantum computation**.

Understanding how noise affects algorithm correctness and efficiency is critical for the future of quantum technologies.

This research therefore seeks to provide insights that contribute toward **more reliable and practical quantum computing systems**.

SLIDE 14 — Thank You

On Slide

Thank

Questions and Comments

You

Speech

Thank you for your attention.

I welcome your questions, comments, and suggestions.